

EOG artifact removal from EEG data: A comparative analysis between MEMD and MSSD

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Abstract—Electroencephalography (EEG) is a tool used in neurological research and clinical diagnostics, but its effectiveness can be hindered by ocular artifacts. This study compares the effectiveness of Multivariate Empirical Mode Decomposition (MEMD) and Canonical Polyadic Decomposition (CPD) in removing these artifacts while preserving neural signal integrity. We use simulated EEG datasets contaminated with ocular artifacts and evaluate their performance with metrics like the Correlation Coefficient (CC) and Relative Root Mean Square Error (RRMSE). Our results show that MEMD generally achieves higher CC values, indicating better preservation of EEG signal integrity, but it has higher variability in RRMSE. In contrast, CPD with frequency grouping shows more consistent and lower RRMSE values. Among CPD grouping methods, frequency grouping outperforms kurtosis and energy grouping in both CC and RRMSE. We also examine how changes in CPD parameters, such as rank, embedding dimension, and grouping method, affect artifact removal. The grouping method has the most significant impact, while rank and embedding dimension also influence performance to a lesser extent. These findings suggest that MEMD is better for preserving signal integrity, whereas CPD with frequency grouping provides more consistent artifact removal. Future research could explore hybrid approaches with CPD, different grouping methods, and optimal parameter settings for various datasets.

I. INTRODUCTION

Electroencephalography (EEG) is a non-invasive diagnostic tool used in neurological, psychological, and neurophysiological research to measure brain activity [19]. The acquisition of EEG data involves placing multiple electrodes on the scalp [22]. While EEG is important in diagnosing conditions such as epilepsy [2], its effectiveness can be compromised by artifacts from various sources, such as ocular movements (EOG short for Electrooculography) and non-physiological sources, including electrode movement and environmental noise. The EEG low-frequency spectrum (0.5–5 Hz) is generally contaminated by eye blink artifacts [6].

EOG artifacts are problematic because eye movements and blinks create electrical potentials that interfere with EEG signals [4]. Traditional methods for removing these artifacts, such as regression algorithms and adaptive filtering, depend on reference signals that may not always be reliable [10]. Advanced techniques like Independent Component Analysis

(ICA) can separate these artifacts without reference channels [16]. However, these methods still have limitations, leading to the loss of valuable information.

Multivariate Empirical Mode Decomposition (MEMD) and Multivariate Singular Spectrum Decomposition (MSSD) [11] are signal processing techniques designed to handle the complexities of multivariate data like EEG. MEMD extends traditional Empirical Mode Decomposition (EMD) to multivariate data, allowing simultaneous decomposition of multiple EEG channels into intrinsic mode functions (IMFs) [20]. MSSD generalizes Singular Spectrum Analysis (SSA) to multivariate settings, decomposing EEG data into interpretable components based on their singular values and vectors. This decomposition isolates meaningful patterns and artifacts, facilitating effective artifact removal.

This study employs both simulated and real EEG datasets to evaluate the performance of MEMD and MSSD. Using metrics such as signal-to-noise ratio (SNR), correlation coefficient (CC), and relative root mean square error (RRMSE), the research aims to determine which method provides better artifact removal without compromising the integrity of the EEG signals.

The questions this paper aims to answer are:

- 1) How do Multivariate Empirical Mode Decomposition (MEMD) and Multivariate Singular Spectrum Decomposition (MSSD) compare in terms of their effectiveness in removing ocular artifacts from EEG signals while preserving the integrity of the underlying neural signals?
- 2) How do the decisions that need to be made for MSSD affect artifact removal capabilities?

To answer these questions, this study will evaluate the performance of MEMD and MSSD on both simulated and real EEG datasets. The analysis will involve testing and comparing the methods using these metrics: CC, and RRMSE. We will also experiment with the different parameters and grouping methods for the MSSD method.

II. RELATED WORK

Various techniques have been developed for ocular artifact (OA) removal in EEG signals, each with distinct approaches and principles.

Regression algorithms use reference electrooculography (EOG) channels to detect and subtract artifacts, relying on

a linear relationship between EOG and EEG signals. These methods depend on the availability and accuracy of reference channels. Adaptive filtering, like the Least Mean Squares (LMS) filter, also uses reference signals to estimate and remove artifacts by continuously adjusting parameters to minimize differences between contaminated EEG and reference signals. However, their performance is tied to the quality of the reference signal [7].

Blind Source Separation (BSS) techniques, such as Independent Component Analysis (ICA), do not require reference signals. ICA decomposes contaminated EEG data into statistically independent components, which can then be used to reconstruct clean EEG signals [9]. Methods to identify artifact components after ICA decomposition include modified multiscale sample entropy (MMSE), kurtosis, and the ICLabel classifier [15]. BSS methods assume statistical independence among sources, which might not always hold true for EEG data. MSSD is also considered a BSS.

Principal Component Analysis (PCA) transforms data into a new coordinate system where principal components capture the largest variance. However, PCA assumes orthogonality, which is not physiologically justified for brain sources [9]. Our approach uses MEMD and MSSD, which do not rely on orthogonality assumptions.

Wavelet Transform methods, including Discrete Wavelet Transform (DWT) and Stationary Wavelet Transform (SWT), decompose signals into different frequency components. These methods are effective for single-channel EEG artifact filtering by isolating artifacts in specific wavelet coefficients and reconstructing the signal after removing these coefficients [5]. They can localize artifacts in both time and frequency domains.

Hybrid approaches combine multiple methods for more effective artifact removal, such as:

Spatially Constrained ICA (SCICA) with Wavelet Denoising (WD) [1]: Combines ICA with wavelet denoising to filter high-frequency artifacts. DWT with Adaptive Noise Cancellation (ANC) [18]: Integrates wavelet decomposition for frequency-domain analysis with adaptive filtering for real-time artifact removal. Empirical Mode Decomposition (EMD) with Canonical Correlation Analysis (CCA) [21]: Uses EMD to decompose signals into intrinsic mode functions (IMFs) and then applies CCA to separate artifacts from neural signals.

MSSD offers an advanced BSS approach that handles multivariate data more effectively. By decomposing EEG data into interpretable components based on singular values and vectors, MSSD isolates meaningful patterns and artifacts, potentially facilitating effective artifact removal and leading to improved signal integrity preservation.

III. BACKGROUND

This section provides an overview of key methods and concepts used in this paper, including Empirical Mode Decomposition (EMD), Singular Spectrum Decomposition (SSD), and the application of tensors for analyzing multi-dimensional data.

A. Empirical Mode Decomposition

Empirical Mode Decomposition (EMD) is a technique for analyzing non-linear and non-stationary signals [8]. The core idea behind EMD is that any complex signal can be represented as a sum of many amplitudes and frequency-modulated zero-mean signals called Intrinsic Mode Functions (IMFs). Each IMF captures different intrinsic oscillatory modes of the data.

The EMD process involves the following steps:

- 1) Identify all the local extrema (both maxima and minima) of the signal $X(t)$.
- 2) Create the upper envelope $e_{\max}(t)$ and lower envelope $e_{\min}(t)$ by interpolating between the local maxima/minima using cubic lines.
- 3) Compute the mean of the upper and lower envelopes:

$$m(t) = \frac{e_{\max}(t) + e_{\min}(t)}{2} \quad (1)$$

- 4) Extract the detail (the difference between the signal and the mean envelope):

$$h(t) = X(t) - m(t) \quad (2)$$

- 5) Check if $h(t)$ meets the criteria for an Intrinsic Mode Function (IMF):
 - a) The number of extrema and zero crossings must either be equal or differ by at most one.
 - b) The mean value of the envelopes defined by the local maxima and minima is zero at every point.
- 6) If $h(t)$ is not an IMF, repeat steps 1-5 on $h(t)$ (sifting process) until $h(t)$ becomes an IMF.
- 7) Once an IMF $c_i(t)$ is obtained, subtract it from the original signal:

$$r(t) = X(t) - c_i(t) \quad (3)$$

- 8) Treat the residue $r(t)$ as the new signal and repeat steps 1-7 to obtain the next IMF.
- 9) Decomposition is interrupted only when the residue has no more extrema.
- 10) The original signal $X(t)$ is finally represented as the sum of the IMFs and the final residue:

$$X(t) = \sum_{i=1}^n c_i(t) + r(t) \quad (4)$$

This iterative process, known as sifting [14], continues until the signal is reduced such that no further IMFs can be extracted.

a) *Multivariate Empirical Mode Decomposition:* Multivariate Empirical Mode Decomposition (MEMD) extends EMD to multivariate data, handling multiple channels simultaneously and ensuring IMFs are aligned across channels. This alignment considers inter-channel correlations and uses a joint decomposition approach [20].

B. Singular Spectrum Decomposition

Singular Spectrum Decomposition (SSD) is a data-driven method for decomposing time series into meaningful components, improving upon Singular Spectrum Analysis (SSA) [3]. SSD operates through embedding, decomposition, component selection, and iteration.

- 1) **Embedding:** Transform the time series $x(n)$ into a Hankel matrix that wraps around the signal, ensuring continuity and capturing the underlying structure:

$$\mathbf{X} = \begin{bmatrix} x_1 & x_2 & \cdots & x_{N-L+1} \\ x_2 & x_3 & \cdots & x_{N-L+2} \\ \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & \cdots & x_N \end{bmatrix}$$

where L is the embedding dimension and N is the length of the original time series.

- 2) **Decomposition:** Apply Singular Value Decomposition (SVD) to $\mathbf{X}$:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

where $\mathbf{U}$ is $L \times L$, $\mathbf{V}$ is $(N-L+1) \times (N-L+1)$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values.

- 3) **Component Selection and Reconstruction:** Analyze the singular values and corresponding singular vectors, and select meaningful components, then reconstruct the time series using diagonal averaging.
- 4) **Iteration:** Repeat until the normalized mean square error (NMSE) falls below a threshold.

SSD automates embedding dimension selection and component grouping, reducing mode mixing and improving interpretability.

a) *Multivariate Singular Spectrum Decomposition:* SSD is originally univariate, making it unsuitable for multi-channel EEG data directly. However, various methods have been developed to extend SSD to multivariate time series. One approach involves creating a Hankel matrix for each channel and stacking them to form a tensor [17]. The approach used in this paper embeds the trajectory matrices of a multivariate series into a tensor.

C. Tensors

Tensors are a generalization of scalars, vectors, and matrices to higher dimensions. They are mathematical objects used to represent multi-dimensional data. A scalar is a zero-order tensor, a vector is a first-order tensor, a matrix is a second-order tensor. Higher-order tensors (third-order and above) extend these concepts into more dimensions.

A tensor can be represented as a multi-dimensional array of numerical values. For example, a third-order tensor $\mathcal{T} \in \mathbb{R}^{I \times J \times K}$ can be represented as:

$$\mathcal{T} = [\mathcal{T}_{ijk}] \quad \text{for } i = 1, \dots, I; \quad j = 1, \dots, J; \quad k = 1, \dots, K.$$

The elements of a tensor are indexed by multiple indices. For a third-order tensor, three indices are required.

Tensors provide a powerful framework for modeling and analyzing multi-dimensional data, making them particularly

useful in fields such as signal processing, machine learning, and neuroscience [13]. In this paper, we utilize tensors to embed the trajectory matrices of a multivariate time series.

IV. ARTIFACT REMOVAL PIPELINE

To address our research questions, two artifact removal frameworks were created: one to evaluate the performance of MEMD and one for MSSD. The following sections outline the methods and algorithms used in each framework.

A. MEMD-DWT

The process of the Multivariate Empirical Mode Decomposition-Discrete Wavelet Transform (MEMD-DWT) pipeline, outlined in Figure 1, involves the following steps:

- 1) **Contaminated EEG:** Start with EEG signals containing ocular artifacts (EOG).
- 2) **Apply MEMD:** Use MEMD to decompose the multichannel EEG signal into Intrinsic Mode Functions (IMFs). The number of directions used was 32.
- 3) **Detect EOG-related IMFs:** Analyze the energy of each IMF to detect those that are likely contaminated with EOG artifacts. The energy of an IMF is computed as the sum of the squared amplitude values. An adaptive thresholding technique with median and interquartile range is used to identify IMFs with significantly high energy levels, which are assumed to be contaminated by EOG artifacts.
- 4) **Apply 5-level DWT:** Apply a 5-level Discrete Wavelet Transform (DWT) to the flagged IMFs, isolating and removing artifact components by zeroing out the approximate coefficients at the final level.
- 5) **Apply inverse MEMD:** Reconstruct the cleaned EEG signal from the modified IMFs using inverse MEMD. IMFs not flagged as contaminated are directly used in the reconstruction process.
- 6) **Cleaned EEG signal:** The output is the cleaned EEG signal with reduced ocular artifacts.
- 7) **RRMSE/CC:** Evaluate the performance of the artifact removal process using the Relative Root Mean Square Error (RRMSE) and the Correlation Coefficient (CC).

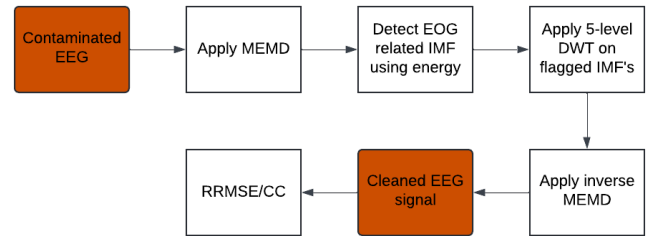


Fig. 1. High level overview of the MEMD-DWT artifact removal pipeline

B. CPD Pipeline

The Canonical Polyadic Decomposition (CPD) pipeline is the approach used for the MSSD part of this research, the methods section will go into more detail on this method. The process is outlined in Figure 3 and involves the following steps:

- 1) **Contaminated EEG:** Start with EEG signals that contain ocular artifacts (EOG).
- 2) **Tensorization:** Convert the multichannel EEG data into a tensor representation. This involves hankelizing the data vectors of each EEG channel with an embedding dimension L of 5. The hankelized matrices of all channels are then reshaped together into a third-order tensor.
- 3) **Apply CPD:** Perform CPD on the tensorized EEG data. CPD decomposes the tensor into a set of factor matrices, isolating the components of the EEG signals into distinct factors.
- 4) **Component selection** The analysis is performed as follows:
 - a) **Power Spectral Density (PSD) Calculation:** The power spectral density of the component is computed using Welch's method on the dehankelized rank-one component obtained from the CPD factor vectors.
 - b) **Frequency Range Selection:** Identify the indices of the frequency range of interest (0.5-5 Hz) [12].
 - c) **Power Calculation:** Calculate the total power within the frequency band and the total power across the entire frequency spectrum.
 - d) **Percentage Power Calculation:** Determine the percentage of power within the frequency band.
 - e) **Artifact Identification:** Identify components as potential artifacts if the percentage of power within the frequency band exceeds a predefined threshold (90% chosen in our research). The reasoning behind this threshold is that ocular artifacts (such as eye blinks and movements) typically dominate the low-frequency range (0.5-5 Hz) in EEG signals. Therefore, if a component has a high percentage of its power concentrated in this frequency band, it is likely to be contaminated by ocular artifacts.
- 5) **Artifact Removal:** Remove the identified artifact components to ensure only clean components are retained.
- 6) **Reconstruct EEG Signal:** Reconstruct the EEG signal from the remaining components. This involves:
 - a) **Combining Components:** Add all the remaining components together to form a final composite component.
 - b) **Dehankelization and Separation:** Convert this composite component back into a tensor by taking the outer product of the three vectors. Dehankelize the tensor slice-by-slice (where each slice is a matrix) to convert it back into a time-series format. This is done by averaging along the anti-diagonals

of the Hankel matrix, effectively reversing the tensorization process.

This can also be seen in Figure 2

- 7) **Cleaned EEG signal:** The output is the cleaned EEG signal with reduced ocular artifacts.
- 8) **RRMSE/CC:** Evaluate the performance of the artifact removal process using the Relative Root Mean Square Error (RRMSE) and the Correlation Coefficient (CC).

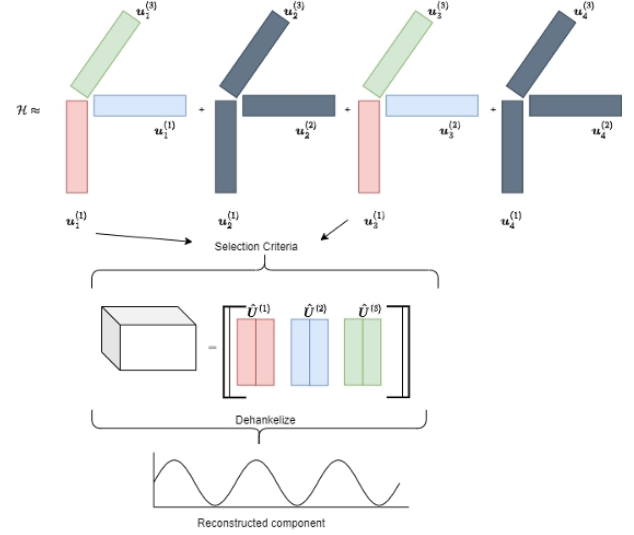


Fig. 2. Overview of the selection and reconstruction process from [11]. Showing the decomposition of a tensor into rank-one components, the selection of relevant components based on specific criteria, the dehankelization of these components, and the final reconstruction of the cleaned EEG signal.

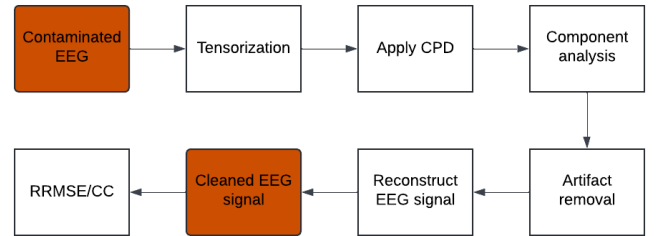


Fig. 3. High-level overview of the CPD artifact removal pipeline

C. Alternative Grouping Methods

In addition to the baseline grouping method used in the CPD pipeline, we also considered two alternative methods for identifying and removing artifact components: grouping by energy and grouping by kurtosis. These methods were evaluated but not selected as the baseline due to their comparative performance.

a) **Grouping by Energy:** The grouping by energy method involves analyzing the energy of each component to identify potential artifacts. The steps are as follows:

- 1) **Energy Calculation:** For each component, the energy is calculated as the sum of the squares of its elements:

$$\text{Energy}(c_i) = \sum_j c_{i,j}^2$$

where c_i is the i -th component.

- 2) **Thresholding:** An adaptive threshold is determined based on the median and interquartile range (IQR) of the energy values:

$$\text{Energy Threshold} = \text{Median Energy} + 1.75 \times \text{IQR Energy}$$

- 3) **Component Selection:** Components with energy below the threshold are considered artifact-free and retained, while those above the threshold are removed.

b) Grouping by Kurtosis: The grouping by kurtosis method focuses on the statistical property of kurtosis to identify artifacts. The steps are as follows:

- 1) **Kurtosis Calculation:** For each component, the kurtosis is calculated as:

$$\text{Kurtosis}(c_i) = \frac{1}{n} \sum_j \left(\frac{c_{i,j} - \mu}{\sigma} \right)^4$$

where μ is the mean, σ is the standard deviation, and n is the number of elements in c_i .

- 2) **Thresholding:** An adaptive threshold is determined based on the median and IQR of the kurtosis values:

$$\text{Kurtosis Threshold} = \text{Median Kurtosis} + 1.5 \times \text{IQR Kurtosis}$$

- 3) **Component Selection:** Components with kurtosis below the threshold are considered artifact-free and retained, while those above the threshold are removed.

These methods were not chosen as the baseline due to their performance compared to the group-by-frequency method. The detailed evaluation of these methods is provided in the results section.

V. METHODS

In this section, we explain the methodologies used for analyzing multivariate time series data, specifically EEG signals. The techniques used include Canonical Polyadic Decomposition (CPD) for tensor decomposition, Multivariate Empirical Mode Decomposition (MEMD) for extracting intrinsic mode functions, and Discrete Wavelet Transform (DWT) for signal decomposition and noise removal. Each method is outlined step-by-step to highlight the process and its application to the EEG data.

A. Canonical Polyadic Decomposition (CPD)

As mentioned in the previous section, for the MSSD part of this research we use Canonical Polyadic Decomposition (CPD), a tensor decomposition technique used to analyze multivariate time series data, such as EEG signals. This method decomposes a tensor into a sum of rank-one tensors, effectively capturing the underlying structure of the data in a more compact and interpretable form. The process of CPD involves several steps, outlined below:

- 1) **Hankelization:** The first step in CPD is transforming the time series data into a Hankel matrix similar to how it is done in Singular spectrum analysis, this is chosen for its simplicity and computational efficiency, making it suitable for handling larger EEG datasets. For a time series $\mathbf{x} = (x_1, x_2, \dots, x_N)$ with an embedding dimension L , the Hankel matrix $\mathbf{H}$ is represented as:

$$\mathbf{H} = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & \cdots & x_N \end{bmatrix}$$

Consider a set of Q time series of length N (these are the different channels). First, each sub-series (channel) is Hankelized according to an embedding dimension L , where $1 < L < N$ and $K = N - L + 1$. After Hankelizing each sub-series, the resulting matrices are stacked sequentially to form the frontal slices of a third-order tensor. In this tensor, the Hankelized matrices occupy the first two modes, while the channel information is represented in the third mode. The resulting tensor, denoted as $\mathcal{H}$ to highlight its Hankel structure, has dimensions $L \times K \times Q$.

- 2) **Embedding Dimension and Rank Selection:** The choice of embedding dimension L and rank R is important for the success of CPD. The embedding dimension L was chosen based on the following formula:

$$L = 1.2 \frac{F_s}{f_{\max}},$$

where F_s is the sampling frequency of the signal and $f_{\max}$ the maximum frequency component of interest in the signal, the rank R of the decomposition represents the number of rank-one tensors that are summed together to approximate the original tensor. The rank R is selected based on the desired level of decomposition. In our study, we used a base-embedding dimension L of 5 and a base-rank R of 16 for the experiments.

- 3) **Tensor Decomposition:** The Canonical Polyadic Decomposition (CPD) algorithm decomposes the tensor into a sum of rank-one tensors. Each rank-one tensor is formed by the outer product of three vectors, which are elements of the factor matrices. The decomposition is given by:

$$\mathcal{X} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r$$

where $\circ$ denotes the outer product, and $\mathbf{a}_r$, $\mathbf{b}_r$, and $\mathbf{c}_r$ are the r -th column vector in the factor matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, for all r from 1 to R (which can also be seen in the first part of 2.) These factor matrices capture the underlying structure of the data across different modes (e.g., time and channels components).

B. Multivariate Empirical Mode Decomposition (MEMD)

Multivariate Empirical Mode Decomposition (MEMD) is an extension of the traditional Empirical Mode Decomposition (EMD) tailored for multivariate time series analysis. MEMD decomposes multivariate signals into a set of intrinsic mode functions (IMFs) that capture different frequency components inherent in the data.

The steps involved in MEMD are as follows:

- 1) **Signal Interpolation:** The multivariate signal is interpolated in multiple directions using uniform angular sampling. This step does not change the time sampling of the original data, which is already evenly sampled. Instead, it ensures that the multichannel data is analyzed from multiple perspectives in a multidimensional space. The number of directions, D , determines how finely the signal is sampled in this space, impacting the accuracy of the decomposition by aligning the Intrinsic Mode Functions (IMFs) across channels.
- 2) **Envelope Mean Calculation:** For each direction, the envelope of the signal is computed. These directions are not the same as channels; instead, they represent various angular perspectives used to analyze the multivariate signal. The envelope mean is then obtained by averaging the envelopes across all directions. This step helps in identifying the local mean of the signal. Mathematically, for a signal $x(t)$ and its envelopes $e_i(t)$ from multiple directions i , the envelope mean $m(t)$ is given by:

$$m(t) = \frac{1}{N} \sum_{i=1}^N e_i(t)$$

where D is the number of directions.

- 3) **IMF Extraction:** The signal is iteratively decomposed into IMFs. This is done by subtracting the envelope mean from the original signal to obtain a residue. This residue is then treated as the new signal, and the process is repeated until the residue becomes a monotonic function or a function with only one extremum. Each IMF $c_k(t)$ is extracted using:

$$c_k(t) = x(t) - m(t)$$

where $x(t)$ is the signal and $m(t)$ is the envelope mean.

- 4) **Stopping Criterion:** The decomposition process stops when a predefined stopping criterion is met. This criterion ensures that the extracted IMFs retain the intrinsic frequency characteristics of the original signal. Common stopping criteria include a predetermined number of IMFs or a threshold on the residue's energy. If $r(t)$ is the residue at any step, the process stops when:

$$\text{Energy}(r(t)) < \epsilon$$

where ϵ is a small threshold value.

Unlike traditional EMD, which is applied to each channel separately, MEMD handles multiple channels simultaneously. This is achieved by considering the joint information across all

channels, thereby ensuring that the IMFs are consistent across the multivariate signal. The number of directions D plays a crucial role in achieving accurate and consistent decomposition across all channels.

The MEMD process can be mathematically represented as:

$$\mathbf{X}(t) = \sum_{k=1}^K \mathbf{C}_k(t) + \mathbf{r}(t)$$

where $\mathbf{X}(t)$ is the multivariate signal, $\mathbf{C}_k(t)$ are the multivariate IMFs, K is the total number of IMFs and $\mathbf{r}(t)$ is the final residue.

C. Discrete Wavelet Transform (DWT)

The Discrete Wavelet Transform (DWT) is a signal processing technique used to decompose a signal into its constituent frequency components. DWT is particularly effective in analyzing non-stationary signals like EEG, as it provides both time and frequency localization. The DWT process involves several key steps, detailed below:

- 1) **Wavelet Decomposition:** The original signal is passed through a series of high-pass and low-pass filters to obtain the detail and approximation coefficients, respectively. Mathematically, if $x[n]$ (the square brackets are used to denote discrete-time signals or sequences) is the original signal, the decomposition can be represented as:

$$A_j[k] = \sum_n x[n]g[2k - n]$$

$$D_j[k] = \sum_n x[n]h[2k - n]$$

where $A_j[k]$ are the approximation coefficients, $D_j[k]$ are the detail coefficients, g is the low-pass filter, and h is the high-pass filter.

- 2) **Multi-Level Decomposition:** The approximation coefficients are further decomposed in subsequent levels to achieve a multi-resolution analysis. This process is repeated iteratively, in our research this is done 5 times, breaking down the signal into various frequency bands. For J levels of decomposition, the signal can be represented as:

$$x[n] = A_J[n] + \sum_{j=1}^J D_j[n]$$

where $A_J[n]$ is the approximation at the J -th level, and $D_j[n]$ are the detail coefficients at the J -th level.

- 3) **Zeroing Approximate Coefficients:** To specifically target and remove low-frequency artifacts, the approximate coefficients $A_j[k]$ are set to zero at the desired decomposition level:

$$A_j[k] = 0 \quad \text{for all } k$$

- 4) **Reconstruction:** The signal is reconstructed by applying the inverse DWT to the detail coefficients. This step

involves up-sampling and filtering the coefficients to reconstruct the signal:

$$x[n] = \sum_k A_J[k]g[2n-k] + \sum_{j=1}^J \sum_k D_j[k]h[2n-k]$$

where g and h are the reconstruction filters corresponding to the low-pass and high-pass filters used in the decomposition.

- 5) **IMF Analysis and Selection:** The IMFs obtained from MEMD are analyzed using DWT to identify and remove noise components. By applying DWT to each IMF, we can effectively isolate and remove high-frequency noise components, preserving the intrinsic structure of the EEG signal.

The DWT pipeline is illustrated in Figure 4.

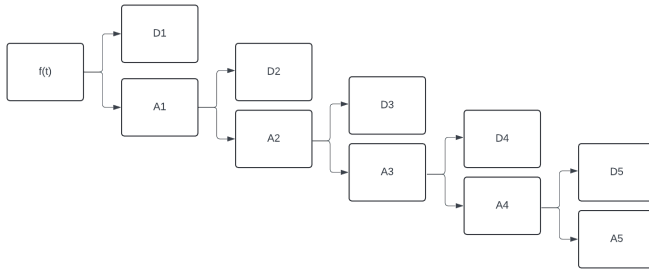


Fig. 4. This figure illustrates the 5-level decomposition of an IMF using DWT. At each level, the signal is split into detail coefficients D_1, D_2, D_3, D_4, D_5 capturing high-frequency content, and approximation coefficients A_1, A_2, A_3, A_4, A_5 capturing low-frequency content. This hierarchical breakdown allows for multi-resolution analysis of the signal's frequency components.

VI. EXPERIMENTAL SETUP

In this section, we describe the experimental setup used to evaluate the effectiveness of our artifact removal techniques. We employ a semi-simulated EEG dataset contaminated with ocular artifacts and assess the performance of the MEMD-DWT and CPD methods using the Relative Root Mean Square Error (RRMSE) and Correlation Coefficient (CC) as metrics. We detail the procedures for applying these methods and analyze their sensitivity to parameter variations.

A. Semi-simulated dataset

The semi-simulated dataset [12] contains the EEG recordings from 27 healthy subjects, which were contaminated manually with OAs. This work randomly considers total 30 EEG signals of 30-second duration, recorded using 19 electrodes, following the 1020 International electrode system standard. For generating the OA contaminated signals, these clean signals were artificially contaminated with the EOG signals from the same subjects following the model:

$$\text{Contaminated_EEG}_{i,j} = \text{Pure_EEG}_{i,j} + a_j \text{VEOG} + b_j \text{HEOG} \quad (5)$$

where Contaminated_EEG is the contaminated EEG data, Pure_EEG is the EEG data without contamination. The VEOG

stands for vertical-EOG, which is the difference between upper and lower EOG electrode signals. Similarly, HEOG represents horizontal-EOG, which is the difference between left and right EOG electrode signals. Vectors a_j and b_j are the contamination coefficients and index i indicates the subject's number, while j denotes the electrode's number.

The dataset consists of artificially contaminated EEG signals, where pure EEG signals were recorded during an eyes-closed session, and EOG signals were used to simulate ocular artifacts. The vertical and horizontal EOG (VEOG and HEOG) signals were combined with the EEG data using linear regression to generate contaminated signals. Contamination coefficients were computed individually for each subject by analyzing blink segments in the VEOG signals and horizontal eye movement in the HEOG signals during eyes-opened sessions [12].

B. Performance Metrics

The effectiveness of the proposed artifact removal techniques is evaluated using the simulated EEG dataset. We use two primary metrics to compare the performance of the MEMD-DWT and CPD methods: the Relative Root Mean Square Error (RRMSE) and the Correlation Coefficient (CC). These metrics provide a quantitative measure of the artifact removal performance and the preservation of the underlying EEG signals.

a) *Relative Root Mean Square Error (RRMSE):* The RRMSE is used to quantify the difference between the ground truth EEG signal (pure EEG) and the cleaned EEG signal obtained after artifact removal. It is defined as:

$$\text{RRMSE} = \sqrt{\frac{\sum_{c=1}^C \sum_{n=1}^N (k_c(n) - \hat{k}_c(n))^2}{\sum_{c=1}^C \sum_{n=1}^N k_c(n)^2}} \times 100\%$$

where $k_c(n)$ is the ground truth EEG signal at the n -th sample in the c -th channel, $\hat{k}_c(n)$ is the cleaned EEG signal at the n -th sample in the c -th channel, N is the number of samples, C is the number of channels, and n is the current sample. A lower RRMSE value indicates better performance, as it signifies that the cleaned EEG signal closely approximates the ground truth signal.

b) *Correlation Coefficient (CC):* The CC is a statistical measure used to assess the degree of similarity between the ground truth EEG signal and the cleaned EEG signal. It is defined as:

$$\text{CC} = \frac{\text{cov}(k, \hat{k})}{\sigma_k \sigma_{\hat{k}}}$$

where $\text{cov}(k, \hat{k})$ is the covariance between the ground truth EEG signal k and the cleaned EEG signal $\hat{k}$, and σ_k and $\sigma_{\hat{k}}$ are the standard deviations of k and $\hat{k}$ respectively. A CC value closer to 1 indicates a high degree of similarity, suggesting that the artifact removal method effectively preserves the original EEG signal's structure.

C. Evaluation Procedure

To assess the performance of the MEMD-DWT and CPD artifact removal methods, we conducted a series of experiments on 30 simulated EEG datasets. For each dataset, we recorded the Root Relative Mean Square Error (RRMSE) and the Correlation Coefficient (CC) as performance metrics.

1) *MEMD-DWT Evaluation*: For the MEMD-DWT method, we applied the decomposition and artifact removal process to all 30 datasets. The resulting RRMSE and CC values were recorded to evaluate the effectiveness of the method in removing ocular artifacts while preserving the underlying neural signals.

2) *CPD Evaluation*: The CPD method was evaluated using baseline values of $R = 16$ and $L = 5$. We used the group-by-frequency method as explained in the artifact removal pipeline with a threshold of 90% to identify and remove artifact components. The RRMSE and CC values were recorded for each of the 30 datasets. Additionally, we tested the alternative grouping methods: group-by-energy and group-by-kurtosis, recording their performance separately on the same 30 datasets. Because the CPD algorithm can get stuck in local minima, it would be best to take the best performance over a number of runs, but due to computation time we run the CPD only once.

3) *Parameter Sensitivity Analysis*: We further investigated the sensitivity of the CPD method to changes in the embedding dimension L and the rank R . Using the group-by-frequency method with a threshold of 90, we experimented with L values ranging from 2 to 12 and R values from 7 to 16 on a single dataset. The baseline configuration of $R = 16$ and $L = 5$ was used for comparison. Performance metrics were recorded to determine the impact of varying these parameters on the effectiveness of the CPD method.

VII. RESULTS

In this section, we present the results of our experimental evaluation of the MEMD-DWT and CPD artifact removal methods. We do this by looking at the CC and RRMSE value's over the different datasets

A. RRMSE Comparison between MEMD and CPD with Frequency Grouping

Figure 5 shows the RRMSE values for MEMD and CPD with frequency grouping. MEMD exhibits higher variability in RRMSE with several peaks, indicating higher errors on specific datasets. But when excluding those peaks it has lower errors in general. In contrast, CPD with frequency grouping maintains lower and more stable RRMSE values, suggesting more consistent performance.

B. RRMSE Comparison among Different CPD Grouping Methods

Figure 6 compares the RRMSE values for CPD using frequency grouping, kurtosis grouping, and energy grouping. CPD with frequency grouping and kurtosis grouping generally perform better than energy grouping, with frequency grouping showing slightly lower RRMSE values overall.

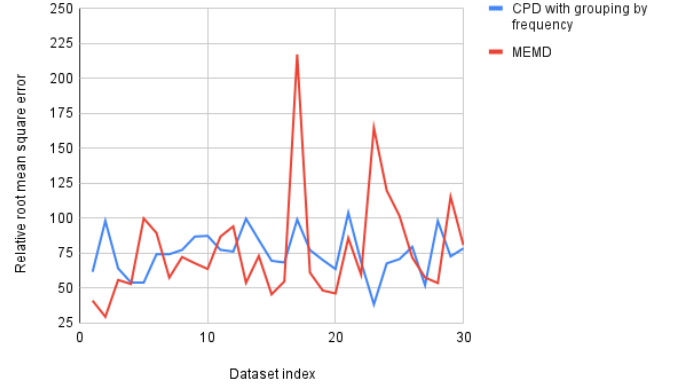


Fig. 5. Comparison of Relative Root Mean Square Error (RRMSE) between MEMD and CPD with Frequency Grouping across 30 datasets.

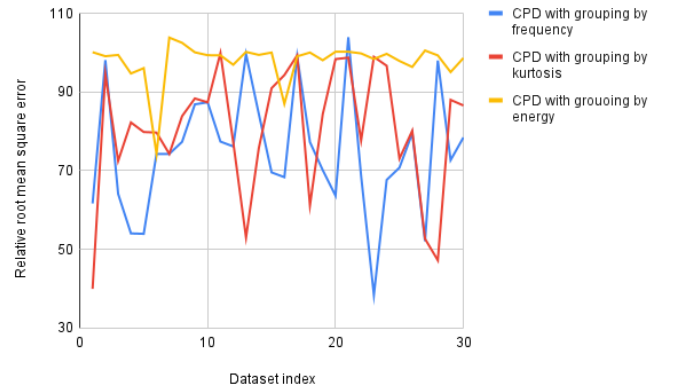


Fig. 6. Comparison of Relative Root Mean Square Error (RRMSE) among CPD with Frequency Grouping, Kurtosis Grouping, and Energy Grouping across 30 datasets.

C. CC Comparison among Different CPD Grouping Methods

Figure 7 presents the correlation coefficients (CC) for CPD with frequency grouping, kurtosis grouping, and energy grouping. Frequency and kurtosis grouping methods maintain higher CC values compared to energy grouping, with frequency grouping showing slightly higher CC values overall.

D. CC Comparison between MEMD and CPD with Frequency Grouping

Figure 8 compares the correlation coefficients (CC) for MEMD and CPD with frequency grouping. Both methods show similar trends, but MEMD generally achieves higher CC values, indicating better preservation of the signal's integrity.

E. Embedding dimension and Rank

The results of the Canonical Polyadic Decomposition (CPD) with varying rank values are presented in Tables I and II. These tables illustrate the performance of the CPD algorithm on two different datasets (23 and 1) in terms of the Correlation Coefficient (CC) and Relative Root Mean Square Error (RRMSE). For dataset 23 we observe a slight improvement in the CC and

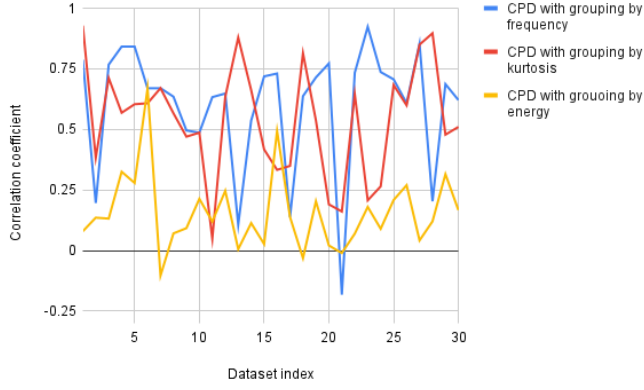


Fig. 7. Comparison of Correlation Coefficient (CC) among CPD with Frequency Grouping, Kurtosis Grouping, and Energy Grouping across 30 datasets.

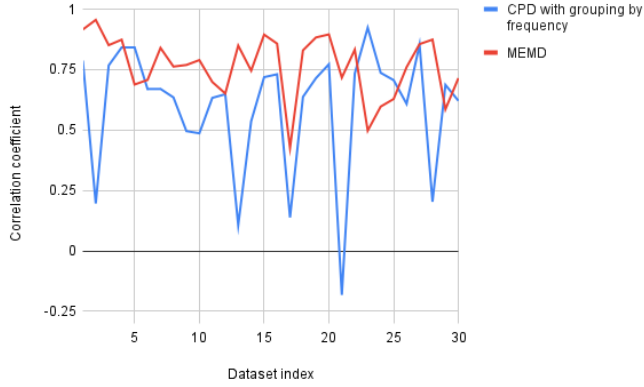


Fig. 8. Comparison of Correlation Coefficient (CC) between MEMD and CPD with Frequency Grouping across 30 datasets.

TABLE I
CC&RRMSE OF DIFFERENT RANKS

Dataset	Rank:	7	8	9	10	11
23	CC	0.9105	0.8137	0.8710	0.9175	0.9191
23	RRMSE	41.5711	59.8078	50.0481	39.8556	39.4708
1	CC	0.7342	0.0687	0.2712	0.0192	0.1135
1	RRMSE	68.7867	99.9801	127.5593	116.3470	99.4558

TABLE II
CC&RRMSE OF DIFFERENT RANKS

Dataset	Rank:	12	13	14	15	16
23	CC	0.9205	0.9215	0.9222	0.9228	0.9233
23	RRMSE	39.1343	38.8953	38.7048	38.5534	38.4303
1	CC	0.2913	0.7801	-0.0046	0.0043	0.3583
1	RRMSE	96.7147	62.6672	107.1509	101.3483	95.5186

RRMSE values when the rank gets higher. Dataset 1 has no noticeable pattern, in general a low performance with a couple of outliers with better performance.

TABLE III
CC&RRMSE OF DIFFERENT EMBEDDING DIMENSIONS

dimension:	2	3	4	5	6	7
RRMSE	37.7952	37.9160	38.1230	38.4303	38.6514	38.8924
CC	0.9258	0.9254	0.9246	0.9233	0.9224	0.9215

TABLE IV
CC&RRMSE OF DIFFERENT EMBEDDING DIMENSIONS

dimension:	8	9	10	11	12
RRMSE	58.0285	39.4343	39.6486	39.8947	40.1842
CC	0.8149	0.9193	0.9184	0.9174	0.9162

The results of the Canonical Polyadic Decomposition (CPD) with varying embedding dimension values are presented in Tables III and IV. These tables illustrate the performance of the CPD algorithm in terms of the Correlation Coefficient (CC) and Relative Root Mean Square Error (RRMSE). We observe that lower values seem slightly better but not by much.

F. Time-series examples

In this section, we present an example of the results of our artifact removal algorithm in the time-series format.

Figure 9 shows the time series plots of the clean and cleaned EEG data for 2500 data points of the first two channels of dataset 2 using CPD and frequency grouping. The top row depicts the clean EEG signals for Channels 1 and 2, while the bottom row shows the corresponding cleaned EEG signals.

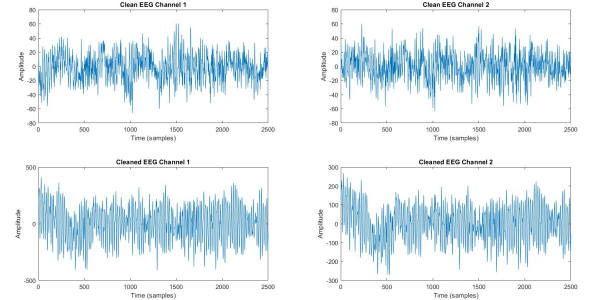


Fig. 9. Time series plots of the clean and cleaned EEG data for Dataset 2. The top row shows the clean EEG signals for Channels 1 and 2. The bottom row shows the corresponding cleaned EEG signals.

Figure 10 shows the time series plots of the clean and cleaned EEG data for 2500 data points of the first two channels of dataset 13 using CPD and energy grouping. The top row depicts the clean EEG signals for Channels 1 and 2, while the bottom row shows the corresponding cleaned EEG signals.

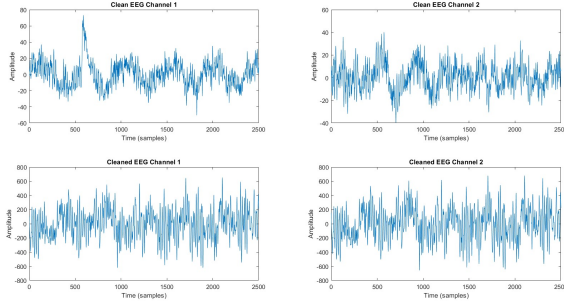


Fig. 10. Time series plots of the clean and cleaned EEG data for Dataset 13 using CPD and group by energy. The top row shows the clean EEG signals for Channels 1 and 2. The bottom row shows the corresponding cleaned EEG signals.

VIII. DISCUSSION

The results presented in the previous section provide a comparison between MEMD-DWT and CPD with various grouping methods in terms of artifact removal performance from EEG data.

A. RRMSE and CC Comparison

a) MEMD-DWT vs. CPD with Frequency Grouping:

As seen in Figure 5, MEMD exhibits higher variability in RRMSE across the 30 datasets, with several peaks indicating higher errors on specific datasets. However, excluding these peaks, MEMD generally maintains lower errors compared to CPD with frequency grouping. On the other hand, CPD with frequency grouping shows more stable and consistently lower RRMSE values overall. This suggests that while MEMD might perform exceptionally well on certain datasets, CPD with frequency grouping offers a more reliable performance across a broader range of datasets.

In terms of the Correlation Coefficient (CC), Figure 8 indicates that MEMD generally achieves higher CC values compared to CPD with frequency grouping. This implies that MEMD is better at preserving the integrity of the EEG signals, maintaining a higher degree of similarity between the cleaned and original signals.

b) *CPD Grouping Methods:* Figures 6 and 7 compare the performance of CPD with different grouping methods: frequency grouping, kurtosis grouping, and energy grouping. CPD with frequency grouping and kurtosis grouping generally perform better than energy grouping. Frequency grouping shows slightly lower RRMSE values and higher CC values overall, indicating its effectiveness in identifying and removing ocular artifacts. As observed in the time-series plots, despite the poor CC and RRMSE values in these cases, the method successfully returns a signal in the same format as the EEG data. However, the significantly higher amplitudes in both examples suggest potential issues with the artifact removal process, indicating a need for further optimization and parameter tuning.

B. Impact of CPD Decision Parameters

The second research question focuses on understanding how the decisions that need to be made for MSSD (i.e., CPD) affect the artifact removal capabilities. The results highlight that:

- 1) **Grouping method** The choice of grouping method significantly impacts the effectiveness of artifact removal. Frequency grouping and kurtosis grouping outperform energy grouping, as evidenced by the lower RRMSE and higher CC values. Frequency grouping, in particular, provides a more reliable identification of artifacts, leading to better overall performance.
- 2) **Rank and embedding dimension** The sensitivity analysis shows that higher ranks generally improve performance, but the effect varies across different datasets. This suggests that the optimal rank may depend on the specific characteristics of the dataset. The embedding dimension has a less pronounced effect, with lower values providing slightly better results. However, the differences are not substantial, suggesting that CPD is relatively robust to changes in the embedding dimension.

IX. CONCLUSION

This study aimed to compare the effectiveness of Multivariate Empirical Mode Decomposition (MEMD) and Canonical Polyadic Decomposition (CPD) in removing ocular artifacts from EEG signals while preserving the integrity of the underlying neural signals. Additionally, it investigated how the decisions regarding CPD impact the artifact removal capabilities.

MEMD was found to be more effective in preserving the signal's integrity, demonstrating higher correlation coefficients overall. However, CPD with frequency grouping exhibited more consistent performance in artifact removal, reflected in lower and more stable RRMSE values. Therefore, the choice between MEMD and CPD should be guided by the specific characteristics of the dataset and the relative importance of signal preservation versus consistency in artifact removal.

Regarding the impact of CPD choices, our study found that different choices such as rank, embedding dimension, and grouping method significantly affect performance. The embedding dimension had the least impact but still required careful selection. The rank of the decomposition played a significant role, with the optimal rank varying across datasets, suggesting it is closely tied to the dataset's characteristics. The choice of grouping method had the most substantial impact on performance, highlighting the importance of selecting an effective grouping strategy for CPD to be successful in artifact removal.

a) *Future Work:* Future research could focus on exploring hybrid approaches that combine the strengths of CPD to achieve superior artifact removal from EEG data. Additionally, further studies could investigate the most effective grouping methods for CPD or establish a correlation between dataset characteristics and the optimal rank. These insights could enhance the adaptability and effectiveness of CPD in various scenarios.

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